

## SAFETY DATA SHEET

Creation Date 16-Jan-2009

Revision Date 28-Mar-2024

Revision Number 3

### 1. Identification

**Product Name** Diethylene glycol dimethyl ether

**Cat No. :** L03840

**CAS No** 111-96-6  
**Synonyms** 2-Methoxyethyl ether; Diethylene glycol dimethyl ether; Diglyme

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Flammable liquids  
Reproductive Toxicity

Category 3  
Category 1B

#### Label Elements

**Signal Word**  
Danger

##### **Hazard Statements**

Flammable liquid and vapor  
May damage fertility. May damage the unborn child

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Use personal protective equipment as required  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Use only non-sparking tools  
Take precautionary measures against static discharge

**Response**

IF exposed or concerned: Get medical attention/advice

**Skin**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store locked up  
Store in a well-ventilated place. Keep cool

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

May form explosive peroxides

### 3. Composition/Information on Ingredients

| Component                        | CAS No   | Weight % |
|----------------------------------|----------|----------|
| Diethylene glycol dimethyl ether | 111-96-6 | <= 100   |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

|                                            |                                                                                                                  |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Most important symptoms and effects</b> | None reasonably foreseeable. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                            |

## 5. Fire-fighting measures

|                                         |                                                                                                                                         |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                |
| <b>Flash Point</b>                      | 51 °C / 123.8 °F                                                                                                                        |
| <b>Method -</b>                         | CC (closed cup)                                                                                                                         |
| <b>Autoignition Temperature</b>         | 170 °C / 338 °F                                                                                                                         |
| <b>Explosion Limits</b>                 |                                                                                                                                         |
| <b>Upper</b>                            | No data available                                                                                                                       |
| <b>Lower</b>                            | No data available                                                                                                                       |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                |

### Specific Hazards Arising from the Chemical

Flammable. May form explosive peroxides. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). peroxides.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 2             | 2                   | 2                  | N/A                     |

## 6. Accidental release measures

|                                  |                                                                                                                                                               |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. |
| <b>Environmental Precautions</b> | Should not be released into the environment.                                                                                                                  |

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Containers should be dated when opened and tested periodically for the presence of peroxides. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals. Keep away from heat, sparks and flame. Store under an inert atmosphere. Protect from moisture. Keep container tightly closed in a dry and well-ventilated place. Incompatible Materials. Strong oxidizing agents. Strong acids. Strong bases. Halogens. Water. |

## 8. Exposure controls / personal protection

### Exposure Guidelines

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas.

### Personal Protective Equipment

#### **Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

#### **Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

#### **Respiratory Protection**

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

#### **Recommended Filter type:**

Organic gases and vapours filter. Type A. Brown. conforming to EN14387.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                              |
|----------------------------------------|------------------------------|
| Physical State                         | Liquid                       |
| Appearance                             | Clear                        |
| Odor                                   | Slight ethereal              |
| Odor Threshold                         | No information available     |
| pH                                     | No information available     |
| Melting Point/Range                    | -64 °C / -83.2 °F            |
| Boiling Point/Range                    | 162 °C / 323.6 °F @ 760 mmHg |
| Flash Point                            | 51 °C / 123.8 °F             |
| Method -                               | CC (closed cup)              |
| Evaporation Rate                       | 0.36 (Butyl acetate = 1.0)   |
| Flammability (solid,gas)               | Not applicable               |
| Flammability or explosive limits       |                              |
| Upper                                  | No data available            |
| Lower                                  | No data available            |
| Vapor Pressure                         | 1.7 mmHg @ 20°C              |
| Vapor Density                          | 4.62 (Air = 1.0)             |
| Specific Gravity                       | 0.937                        |
| Solubility                             | Miscible with water          |
| Partition coefficient; n-octanol/water | No data available            |
| Autoignition Temperature               | 170 °C / 338 °F              |
| Decomposition Temperature              | No information available     |
| Viscosity                              | 2 cP at 20 °C                |
| Molecular Formula                      | C6 H14 O3                    |
| Molecular Weight                       | 134.18                       |

## 10. Stability and reactivity

### **Reactive Hazard**

None known, based on information available

### **Stability**

Hygroscopic. May form explosive peroxides.

|                                         |                                                                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. Exposure to moist air or water. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Strong acids, Strong bases, Halogens, Water                                                                  |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), peroxides                                                                    |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                              |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                                         |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Component Information

| Component                        | LD50 Oral                 | LD50 Dermal | LC50 Inhalation                            |
|----------------------------------|---------------------------|-------------|--------------------------------------------|
| Diethylene glycol dimethyl ether | LD50 = 7500 mg/kg ( Rat ) | Not listed  | LC50 > 11000 mg/m <sup>3</sup> ( Rat ) 7 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                        | CAS No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------------------|----------|------------|------------|------------|------------|------------|
| Diethylene glycol dimethyl ether | 111-96-6 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** Mutagenic effects have occurred in experimental animals.

**Reproductive Effects** Experiments have shown reproductive toxicity effects on laboratory animals.

**Developmental Effects** Developmental effects have occurred in experimental animals.

**Teratogenicity** Teratogenic effects have occurred in experimental animals.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

| Component                        | Freshwater Algae | Freshwater Fish                     | Microtox                      | Water Flea |
|----------------------------------|------------------|-------------------------------------|-------------------------------|------------|
| Diethylene glycol dimethyl ether | Not listed       | Rainbow trout: LC50 = 9845 mg/L/96h | Daphnia: EC50 = 5868 mg/L/96h | Not listed |

|  |  |                                                                                       |  |  |
|--|--|---------------------------------------------------------------------------------------|--|--|
|  |  | Fathaed Minnow: LC50 =<br>8569 mg/L 96h<br>Bluegill/Sunfish: LC50 =<br>10928 mg/L 96h |  |  |
|--|--|---------------------------------------------------------------------------------------|--|--|

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** . Will likely be mobile in the environment due to its water solubility.

| Component                        | log Pow |
|----------------------------------|---------|
| Diethylene glycol dimethyl ether | -0.36   |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN3271  
 Proper Shipping Name ETHERS, N.O.S.  
 Hazard Class 3  
 Packing Group III

#### TDG

UN-No UN3271  
 Proper Shipping Name ETHERS, N.O.S.  
 Hazard Class 3  
 Packing Group III

#### IATA

UN-No UN3271  
 Proper Shipping Name ETHERS, N.O.S.\*  
 Hazard Class 3  
 Packing Group III

#### IMDG/IMO

UN-No UN3271  
 Proper Shipping Name ETHERS, N.O.S.  
 Hazard Class 3  
 Packing Group III

### 15. Regulatory information

#### United States of America Inventory

| Component                        | CAS No   | TSCA | TSCA Inventory notification -<br>Active-Inactive | TSCA - EPA Regulatory<br>Flags |
|----------------------------------|----------|------|--------------------------------------------------|--------------------------------|
| Diethylene glycol dimethyl ether | 111-96-6 | X    | ACTIVE                                           | S                              |

#### **Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

S - Indicates a substance that is identified in a proposed or final Significant New Use Rule.

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)** Not applicable

#### **TSCA 12(b)** - Notices of Export

| Component                        | CAS No   | TSCA 12(b) - Notices of Export |
|----------------------------------|----------|--------------------------------|
| Diethylene glycol dimethyl ether | 111-96-6 | Section 5                      |

**International Inventories**

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                        | CAS No   | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|----------------------------------|----------|-----|------|-----------|-------|------|------|------|-------|----------|
| Diethylene glycol dimethyl ether | 111-96-6 | X   | -    | 203-924-4 | X     | X    | X    | X    | X     | KE-27705 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component                        | CAS No   | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|----------------------------------|----------|----------|-------------------------------|---------------------------------|
| Diethylene glycol dimethyl ether | 111-96-6 | <= 100   | 1.0 %                         | -                               |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)** Not applicable

**Clean Air Act**

| Component                        | HAPS Data | Class 1 Ozone Depletors | Class 2 Ozone Depletors |
|----------------------------------|-----------|-------------------------|-------------------------|
| Diethylene glycol dimethyl ether | X         |                         | -                       |

**OSHA - Occupational Safety and Health Administration** Not applicable

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component                        | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|----------------------------------|---------------|------------|--------------|----------|--------------|
| Diethylene glycol dimethyl ether | -             | X          | X            | X        | -            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

**Other International Regulations**

Mexico - Grade

Moderate risk, Grade 2

## Authorisation/Restrictions according to EU REACH

| Component                        | CAS No   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization                                                                          | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Diethylene glycol dimethyl ether | 111-96-6 | Toxic for reproduction<br>Category 1B, Article 57<br>Application date: February 22, 2016<br>Sunset date: August 22, 2017<br>Exemption - None | Use restricted. See item 30.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | SVHC Candidate list - Toxic for reproduction (Article 57c)                                            |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

## REACH links

<https://echa.europa.eu/authorisation-list>
<https://echa.europa.eu/substances-restricted-under-reach>
<https://echa.europa.eu/candidate-list-table>

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                        | CAS No   | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------------------------|----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Diethylene glycol dimethyl ether | 111-96-6 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

## Contains component(s) that meet a 'definition' of per &amp; poly fluoroalkyl substance (PFAS)?

Not applicable

## Other International Regulations

| Component                        | CAS No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Diethylene glycol dimethyl ether | 111-96-6 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

## Prepared By

Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

## Creation Date

16-Jan-2009

## Revision Date

28-Mar-2024

## Print Date

28-Mar-2024

## Revision Summary

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**